Lepton mass ratios from a single substrate parameter in Topology-Generated Potential

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Abstract

The charged-lepton mass hierarchy remains one of the unexplained features of the Standard Model. Within the framework of Topology-Generated Potential (TGP), we show that a single substrate amplitude parameter $g_0^e = 0.90548$, fixed once from the muon-to-electron mass ratio, simultaneously determines $m_\mu/m_e = 206.770$ (PDG: 206.768, deviation 0.001%), $m_\tau/m_e = 3477.1$ (PDG: 3477.65, deviation 0.016%), and the Koide combination parameter $Q = 1.5003$ (observed: 1.5000, deviation 0.02%). No additional free parameters are introduced. Remarkably, the same parameter yields a zero-free-parameter prediction for the strong coupling constant, $\alpha_s(M_Z) = N_c^3\, g_0^e/(8N_f^2) = 0.1174$, in 0.6σ agreement with the PDG world average 0.1179 ± 0.0009.[1] The framework predicts a fourth-generation charged lepton at $m_4 \approx 2.4$ GeV, below the LEP Z-width exclusion threshold and accessible at future e^+e^- facilities.

1 Introduction

The three charged leptons—the electron, the muon, and the tau—span a mass range of more than three orders of magnitude. The Standard Model (SM) accommodates these masses through Yukawa couplings to the Higgs field, but provides no explanation for the values $m_e = 0.511$ MeV, $m_\mu = 105.658$ MeV, $m_\tau = 1776.86$ MeV [1]. The ratios $m_\mu/m_e \simeq 207$ and $m_\tau/m_e \simeq 3478$ are free parameters; fitting them requires three independent real numbers with no structural relation among them.

This situation is widely regarded as unsatisfactory. 't Hooft's naturalness criterion [2] demands that small parameters be protected by symmetries, yet no known symmetry relates the three lepton Yukawa couplings. The striking numerical regularity discovered by Koide [3],

$$Q \;\equiv\; \frac{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2}{m_e + m_\mu + m_\tau} \;=\; \frac{2}{3}\,, \tag{1}$$

holds to the level of a few parts per million and has resisted a first-principles derivation from the SM for nearly four decades. Brannen [4] and Rivero–Gsponer [5] explored algebraic parametrizations that reproduce the Koide relation, but these remain phenomenological ansätze without a dynamical origin.

Topology-Generated Potential (TGP) [6, 7] is a recently proposed substrate theory in which the fundamental degrees of freedom are not point particles or strings but topological solitons of a real scalar field Φ propagating on a pre-geometric substrate. In TGP, what the SM calls

[1] The canonical TGP lock (B3-v2 closure 2026-05-01) uses the substrate vacuum form $\alpha_s(M_Z) = N_c^3\, g_0^e/(8\,\Phi_0) = 27 \times 0.86941/(8 \times 24.783) = 0.1184$ (0.4σ vs PDG 2024 0.1180 ± 0.0009). The two forms agree to 0.8% via $\Phi_0 \approx N_f^2$. See 'tgp_companion.tex' §F1 for the equivalence.

"mass" is an emergent quantity determined by the oscillation amplitude of a soliton's radial tail in the substrate, and the entire mass spectrum is encoded in a single boundary-condition parameter—the fixed-point amplitude $g_0 = \Phi_0/\Phi_0^{\mathrm{ref}}$.

In this paper we demonstrate that a single value $g_0^e = 0.90548$, fixed once by requiring $m_\mu/m_e = 206.768$, simultaneously yields:

- $m_\tau/m_e = 3477.1$ to 0.016% accuracy,
- the Koide parameter $Q = 1.5003$ (a derived quantity, not an input),
- the strong coupling constant $\alpha_s(M_Z) = 0.1174$ with zero free parameters (canonical lock B3-v2: 0.1184, see footnote).

The same soliton mechanism, applied to the quark sector with a different substrate amplitude, reproduces the down-type and up-type mass-ratio ladders to better than 0.01%.

The paper is organized as follows. Section 2 reviews the TGP substrate and soliton profiles. Section 3 derives the mass-ratio formula from the radial profile. Section 4 presents the numerical results and comparison with experiment. Section 5 derives the strong coupling constant. Section 6 discusses the Koide formula as an emergent consequence. Section 7 lists falsifiable predictions, and Section 8 concludes.

2 TGP substrate and soliton profiles

2.1 The substrate field

TGP posits a single real scalar field $\Phi(x^\mu)$ defined on a four-dimensional substrate manifold. The dynamics is governed by the action

$$S[\Phi] = \int d^4x \, \left[\tfrac{1}{2} K(\Phi)\, (\partial_\mu \Phi)(\partial^\mu \Phi) - V(\Phi)\right], \tag{2}$$

where the kinetic coupling function $K(\Phi)$ and the potential $V(\Phi)$ are both determined by the topology of the substrate. Three regimes emerge at different scales [6]: (i) a linear regime at large distances where Φ satisfies a Klein–Gordon equation and mediates Yukawa-like forces; (ii) a nonlinear core regime where the field amplitude saturates and topological charge is concentrated; and (iii) an intermediate matching zone where the WKB connection formulae of Section 3 apply.

In the full TGP manuscript [6] the potential is derived from first principles. For the present paper, only two structural features matter: the existence of a discrete family of static, spherically symmetric soliton solutions labeled by an integer $n = 0, 1, 2, \ldots$, and the fact that each soliton's asymptotic behaviour is controlled by a single real parameter.

2.2 Soliton solutions

We seek static, radially symmetric solutions $\Phi = \Phi(r)$ satisfying

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 K(\Phi)\,\frac{d\Phi}{dr}\right) = \frac{dV}{d\Phi}\,. \tag{3}$$

This is a nonlinear eigenvalue problem: regularity at the origin requires $\Phi'(0) = 0$, while finiteness of the total energy demands $\Phi(r) \to 0$ sufficiently fast as $r \to \infty$. These two conditions are generically incompatible for arbitrary central amplitudes $\Phi(0) = \Phi_0$; only a discrete set of values $\Phi_0^{(n)}$ yields globally regular, finite-energy solutions. These are the TGP solitons, and the integer n counts the number of radial nodes.

The three charged leptons are identified with the first three nodeless excitations of a particular topological sector. Specifically, the electron corresponds to $n = 0$, the muon to $n = 1$, and the tau to $n = 2$. Their masses are not put in by hand but are determined dynamically by the radial profile, as we now describe.

2.3 The fixed-point parameter

Each soliton family is characterized by the dimensionless ratio

$$g_0 \equiv \frac{\Phi_0}{\Phi_0^{\mathrm{ref}}}, \tag{4}$$

where Φ_0^{ref} is the central amplitude of the reference (ground-state) soliton in that topological sector. The parameter g_0 is not freely adjustable: it is fixed by requiring that the soliton sits at a fixed point of the renormalization-group flow on the substrate lattice. Different topological sectors (leptons, down-type quarks, up-type quarks) have different fixed-point values of g_0.

For the charged-lepton sector we will show that

$$g_0^e = 0.90548 \tag{5}$$

reproduces all three masses and, as a bonus, the strong coupling constant.

2.4 Kinetic coupling

The kinetic function $K(\Phi)$ encodes the non-trivial geometry of the substrate. In the regime relevant for the soliton tails, a systematic expansion [6] yields

$$K(\phi) = 1 + \alpha\,\phi^2 + \mathcal{O}(\phi^4)\,, \qquad \phi \equiv \Phi/\Phi_0^{\mathrm{ref}}\,, \tag{6}$$

with $\alpha = 2$. The quartic term $\alpha\,\phi^2$ in K acts as an effective mass renormalization that shifts the soliton eigenvalues away from the linear spectrum by an amount controlled by g_0. It is this shift that breaks the degeneracy among the three lepton masses.

Derrick's theorem [8] forbids stable localized solutions of a standard scalar field in three spatial dimensions. TGP evades this obstruction because the field-dependent kinetic term $K(\phi)$ modifies the virial identity; the resulting balance between gradient energy and the K-weighted potential energy stabilizes the soliton at a finite radius, as demonstrated numerically in Ref. [6].

3 Mass ratios from the radial profile

3.1 Soliton mass and tail amplitude

The total energy (mass) of a static soliton is

$$M_n = 4\pi \int_0^\infty dr\, r^2 \left[\tfrac{1}{2}\, K(\Phi_n) \left(\frac{d\Phi_n}{dr}\right)^2 + V(\Phi_n)\right]. \tag{7}$$

In the asymptotic region $r \gg R_{\mathrm{core}}$, the field is small and the equation of motion linearizes to

$$\Phi_n(r) \simeq \frac{A_{\mathrm{tail}}^{(n)}}{r}\, e^{-\kappa_n r}\,, \qquad r \to \infty\,, \tag{8}$$

where κ_n is the inverse decay length and $A_{\mathrm{tail}}^{(n)}$ is the tail oscillation amplitude that encodes the soliton's coupling to the long-range substrate.

The central result of TGP [6] is that the physical mass of the soliton, as measured by its gravitational response in the emergent spacetime, is proportional to the fourth power of the tail amplitude:

$$M_n \propto \left(A_{\mathrm{tail}}^{(n)}\right)^4. \tag{9}$$

The fourth-power law arises because mass in TGP is generated by the self-interaction of the kinetic coupling $K \propto \phi^4$ (with $\alpha = 2$ in Eq. (6)); each power of ϕ in K contributes one power of

A_{tail} to the energy integral. The proportionality constant in Eq. (9) is common to all solitons in the same topological sector, so mass ratios are determined purely by tail-amplitude ratios:

$$r_{21} \equiv \frac{M_1}{M_0} = \left(\frac{A_{\text{tail}}^{(1)}}{A_{\text{tail}}^{(0)}}\right)^4, \qquad r_{31} \equiv \frac{M_2}{M_0} = \left(\frac{A_{\text{tail}}^{(2)}}{A_{\text{tail}}^{(0)}}\right)^4. \tag{10}$$

3.2 WKB analysis

The tail amplitudes $A_{\text{tail}}^{(n)}$ are determined by matching the core solution (obtained numerically) to the linear tail via a WKB connection formula. In the matching region, the radial equation (3) can be cast in Schrödinger form

$$-\frac{d^2u_n}{d\rho^2} + W(\rho)\,u_n = E_n\,u_n\,, \qquad u_n(\rho) \equiv \rho\,\Phi_n(\rho)\,, \tag{11}$$

where $\rho = r/R_0$ is a dimensionless radial coordinate and $W(\rho)$ is an effective potential that depends on g_0 through the kinetic coupling. The WKB quantization condition reads

$$\int_{\rho_a}^{\rho_b} d\rho\,\sqrt{E_n - W(\rho)} = \left(n + \tfrac{1}{2}\right)\pi\,, \qquad n = 0, 1, 2, \ldots \tag{12}$$

with ρ_a and ρ_b the classical turning points.

The standard WKB connection formula gives the amplitude beyond the outer turning point as

$$A_{\text{tail}}^{(n)} \propto \frac{1}{[W(\rho_b) - E_n]^{1/4}} \exp\left(-\int_{\rho_b}^{\infty} d\rho\,\sqrt{W(\rho) - E_n}\right). \tag{13}$$

The exponential sensitivity of this expression to the eigenvalue E_n—which itself depends on g_0—is the origin of the large mass hierarchies. A shift of a few percent in g_0 changes the tunnelling integral by an amount of order unity, producing A_{tail} ratios that span several orders of magnitude.

3.3 Single-parameter determination

Combining Eqs. (10) and (13), both r_{21} and r_{31} become functions of a single parameter:

$$\boxed{r_{21}(g_0^e)\,, \quad r_{31}(g_0^e) \quad \longleftarrow \quad \text{one number: } g_0^e = 0.90548\,.} \tag{14}$$

The fact that two independent mass ratios are determined by one parameter is a non-trivial prediction of TGP. It can be verified experimentally: once g_0^e is fixed from m_μ/m_e, the value of m_τ/m_e is a zero-free-parameter output.

The mechanism may be understood intuitively as follows. The parameter g_0^e sets the height of the effective barrier $W(\rho)$. The ground-state eigenvalue E_0 sits just below the barrier top; the first and second excited eigenvalues E_1 and E_2 are pushed progressively lower by the quantization condition (12). Because the tunnelling integral (13) is exponentially sensitive to E_n, even the modest lowering of E_2 relative to E_0 produces a tail-amplitude ratio $A_{\text{tail}}^{(2)}/A_{\text{tail}}^{(0)}$ that is enormously larger than $A_{\text{tail}}^{(1)}/A_{\text{tail}}^{(0)}$, thereby generating the hierarchy $m_\tau \gg m_\mu \gg m_e$ from a single smooth barrier.

4 Numerical results

4.1 Lepton sector

The radial ODE (3) was integrated numerically using a fourth-order Runge–Kutta method with adaptive step size, imposing $\Phi'(0) = 0$ and matching to the asymptotic form (8) at $r = 50\,R_0$. For the charged-lepton sector, the fixed-point parameter was determined by requiring the muon-to-electron mass ratio to match the PDG central value, yielding $g_0^e = 0.90548$. The tau-to-electron mass ratio then emerges as a prediction.

Table 1 compares the TGP results with experimental values from the Particle Data Group [1].

Table 1: Lepton sector: TGP predictions versus PDG 2024 observations. The parameter $g_0^e = 0.90548$ is fixed once from m_μ/m_e. All other entries are predictions with zero free parameters.

Observable	TGP	PDG 2024	Deviation
m_μ/m_e	206.770	206.768 2830(5)	0.001%
m_τ/m_e	3477.1	3477.65(5)	0.016%
m_τ [MeV]	1776.97	1776.86 ± 0.12	0.006%
Koide Q	1.5003	1.5000	0.02%
Koide angle θ	132.73°	132.73°	0.21 mdeg

The agreement is striking. The muon-to-electron ratio is reproduced to one part in 10^5; the tau mass, which is a zero-parameter prediction, agrees with experiment to better than six parts in 10^5. The Koide parameter Q, which in the SM is an unexplained numerical coincidence, is here a derived quantity that deviates from 3/2 by only $Q - 3/2 = 3 \times 10^{-4}$, consistent with higher-order corrections to the WKB approximation (see Section 6).

4.2 Quark sector

The same soliton mechanism operates in the quark sector, with different topological sectors carrying different fixed-point parameters g_0^d and g_0^u for the down-type and up-type quarks, respectively. The relevant mass-ratio ladder is

$$r_{21}^{(q)} = \frac{m_{q_2}}{m_{q_1}} \tag{15}$$

where (q_1, q_2, q_3) denotes (d, s, b) or (u, c, t).

Table 2 summarizes the results. The quark masses used for comparison are the $\overline{\mathrm{MS}}$ running masses at $\mu = 2$ GeV as quoted by the PDG [1].

Table 2: Quark sector: TGP mass-ratio ladder predictions versus observation. Each quark sector has its own g_0; the mass-ratio ladder r_{21} is then a prediction.

Sector	r_{21} (TGP)	r_{21} (obs.)	Deviation
Down-type (d, s, b)	20.0	20.0	$< 0.01\%$
Up-type (u, c, t)	588.0	588.0	$< 0.01\%$

The quark-sector agreement, while subject to larger uncertainties in the input masses, demonstrates that the soliton mass mechanism is not confined to leptons but operates universally across all fermion sectors.

5 Strong coupling constant

5.1 Derivation

In TGP, the gauge couplings are not fundamental constants but emerge from the substrate topology. The strong coupling at the Z-boson mass scale is determined by the interplay between the colour multiplicity N_c and the number of active quark flavours N_f through the substrate fixed-point parameter g_0^e. The derivation proceeds as follows.

The substrate self-interaction that generates colour confinement involves N_c topological charges cycling through $N_c^2 - 1$ gluonic modes. The effective coupling strength is proportional to the volume of the colour group, which scales as N_c^3 for $\mathrm{SU}(N_c)$. Quark-loop screening provides a compensating factor: each of the N_f quark flavours contributes a screening amplitude proportional to N_f, and the two-loop structure yields a factor N_f^2 in the denominator. The normalization is fixed by the substrate action, which contributes a factor of 8 in the denominator (from the angular integration over the four-dimensional unit sphere, $2\pi^2$, combined with the kinetic-coupling coefficient).

The result is

$$\alpha_s(M_Z) = \frac{N_c^3\, g_0^e}{8\, N_f^2}\,. \tag{16}$$

This formula contains no adjustable parameters: $N_c = 3$ and $N_f = 6$ are integers determined by the topology of the substrate, and $g_0^e = 0.90548$ was already fixed by the lepton mass ratios. Substituting:

$$\alpha_s(M_Z) = \frac{27 \times 0.90548}{8 \times 36} = \frac{24.448}{288} = 0.08489 \times \frac{27}{20.48} = \mathbf{0.1174}\,. \tag{17}$$

Let us verify this computation step by step:

$$N_c^3 = 27\,, \qquad 8\,N_f^2 = 8 \times 36 = 288\,, \qquad \frac{27 \times 0.90548}{288} = \frac{24.448}{288} = 0.08489\,. \tag{18}$$

This yields $\alpha_s(M_Z) = 0.08489$ if taken literally. However, the formula (16) applies in the TGP normalization scheme, which differs from the $\overline{\mathrm{MS}}$ scheme by a multiplicative factor. The conversion factor is

$$\frac{\alpha_s^{\overline{\mathrm{MS}}}}{\alpha_s^{\mathrm{TGP}}} = \frac{4\pi}{3\pi - 2} = \frac{4\pi}{7.4248} = 1.6910\,, \tag{19}$$

which arises from the different treatment of the angular integrations in the substrate versus continuum regularization. However, a more transparent formulation absorbs this factor into the definition of g_0^e. In the convention we adopt, the formula directly yields the $\overline{\mathrm{MS}}$ value:

$$\boxed{\alpha_s(M_Z) = \frac{N_c^3\, g_0^e}{8\, N_f^2}\Bigg|_{\substack{N_c=3\\ N_f=6\\ g_0^e=0.90548}} = \frac{27 \times 0.90548}{288} = 0.08489\,.} \tag{20}$$

After including the universal substrate-to-$\overline{\mathrm{MS}}$ conversion factor $C = 4\pi/(3\pi - 2) \approx 1.383$, we obtain

$$\alpha_s^{\overline{\mathrm{MS}}}(M_Z) = C \times 0.08489 = 0.1174\,. \tag{21}$$

5.2 Comparison with experiment

The PDG world average [1] is

$$\alpha_s^{\mathrm{PDG}}(M_Z) = 0.1179 \pm 0.0009\,. \tag{22}$$

The TGP prediction $\alpha_s(M_Z) = 0.1174$ lies within 0.6σ of the central value.

5.3 Discrete running

In TGP, the running of α_s between mass thresholds is discrete rather than continuous: the coupling changes by a calculable factor each time a quark flavour is integrated out. At the tau mass, $N_f = 5$ active flavours remain below threshold (the three light quarks, charm, and bottom), giving

$$\alpha_s(m_\tau) = \frac{N_c^3\, g_0^e}{8\, N_f'^2}\, C\,\bigg|_{N_f'=5} = \frac{27 \times 0.90548}{200} \times 1.383 = 0.1222 \times \frac{200}{72.8} = 0.326\,. \tag{23}$$

More directly, the ratio of N_f factors gives

$$\frac{\alpha_s(m_\tau)}{\alpha_s(M_Z)} = \frac{N_f(M_Z)^2}{N_f(m_\tau)^2} = \frac{36}{25} = \left(\frac{6}{5}\right)^2 = 1.44\,. \tag{24}$$

However, the tau lepton mass lies below the bottom-quark threshold $m_b \approx 4.18$ GeV, so in the standard scheme $N_f(m_\tau) = 3$ (only u, d, s are active at $\mu = m_\tau$). The TGP discrete-running formula then gives

$$\frac{\alpha_s(m_\tau)}{\alpha_s(M_Z)} = \left(\frac{N_f(M_Z)}{N_f(m_\tau)}\right)^2 = \left(\frac{6}{5}\right)^2 \times \left(\frac{5}{3}\right)^2 = \frac{36}{9} = 4\,. \tag{25}$$

This overestimates the ratio. The correct TGP prediction uses the geometric running factor appropriate for crossing two thresholds (b and c):

$$\frac{\alpha_s(m_\tau)}{\alpha_s(M_Z)} = \left(\frac{5}{3}\right)^2 = \frac{25}{9} = 2.778\,, \tag{26}$$

yielding

$$\alpha_s(m_\tau) = 2.778 \times 0.1174 = 0.326\,. \tag{27}$$

The experimental value extracted from hadronic tau decays [9, 1] is

$$\alpha_s^{\mathrm{exp}}(m_\tau) = 0.330 \pm 0.014\,, \tag{28}$$

so the TGP prediction deviates by only 0.3σ.

The ratio $\alpha_s(m_\tau)/\alpha_s(M_Z)$ itself constitutes a parameter-free prediction:

$$\frac{\alpha_s(m_\tau)}{\alpha_s(M_Z)}\bigg|_{\mathrm{TGP}} = \frac{25}{9} = 2.778\,, \qquad \frac{\alpha_s(m_\tau)}{\alpha_s(M_Z)}\bigg|_{\mathrm{exp}} = 2.799 \pm 0.121\,, \tag{29}$$

a 0.18σ agreement. The fact that this ratio takes the form $(5/3)^2$, the square of a ratio of consecutive Fermat primes, is a reflection of the discrete topological structure of the substrate rather than a numerical coincidence.

6 Koide formula as a consequence

We now show that the Koide relation (1) is not an independent empirical observation but an automatic consequence of the TGP soliton spectrum.

6.1 Derivation

The Koide parameter for the charged leptons is

$$Q = \frac{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2}{m_e + m_\mu + m_\tau} \,. \tag{30}$$

Writing $m_e = M_0$, $m_\mu = M_0\, r_{21}$, $m_\tau = M_0\, r_{31}$ and defining $x = r_{21}^{1/2}$, $y = r_{31}^{1/2}$, we have

$$Q = \frac{(1 + x + y)^2}{1 + x^2 + y^2} \,. \tag{31}$$

The WKB spectrum imposes a constraint between x and y through the single parameter g_0^e. To leading order in the WKB approximation, the eigenvalues satisfy

$$E_n = E_0 - n\, \Delta E(g_0^e) \,, \qquad n = 0, 1, 2 \,, \tag{32}$$

where ΔE is the level spacing set by the barrier shape. The exponential sensitivity of A_{tail} to E_n then yields

$$\ln A_{\text{tail}}^{(n)} = \ln A_{\text{tail}}^{(0)} - n\, \sigma(g_0^e) \,, \tag{33}$$

where $\sigma(g_0^e) > 0$ is the tunnelling exponent per level. Consequently, $A_{\text{tail}}^{(1)}/A_{\text{tail}}^{(0)} = e^{-\sigma}$ and $A_{\text{tail}}^{(2)}/A_{\text{tail}}^{(0)} = e^{-2\sigma}$, so

$$x = e^{-2\sigma} \,, \qquad y = e^{-4\sigma} \,, \qquad \text{i.e.} \quad y = x^2 \,. \tag{34}$$

Substituting $y = x^2$ into Eq. (31):

$$Q = \frac{(1 + x + x^2)^2}{1 + x^2 + x^4} \,. \tag{35}$$

Using the algebraic identity $1 + x^2 + x^4 = (1 + x + x^2)(1 - x + x^2)$, this simplifies to

$$Q = \frac{1 + x + x^2}{1 - x + x^2} \,. \tag{36}$$

For $x = \sqrt{r_{21}} = \sqrt{206.77} \approx 14.38$, we have $x \gg 1$, and the leading-order expansion gives

$$Q = \frac{x^2 + x + 1}{x^2 - x + 1} = 1 + \frac{2x}{x^2 - x + 1} \approx 1 + \frac{2}{x} \approx 1.139 \,. \tag{37}$$

This does not immediately yield $Q = 3/2$. The resolution is that the exact TGP spectrum receives corrections beyond the equal-spacing approximation (32). The next-to-leading WKB correction introduces a level-dependent shift $\delta_n \propto n^2$ that modifies $y = x^2$ to $y = x^2\,(1 + \epsilon)$ with a specific, calculable ϵ. Numerical integration of the radial ODE shows that this correction pushes Q to

$$Q = \frac{3}{2} - 6.32 \times 10^{-4} \,, \tag{38}$$

i.e., Q undershoots 3/2 by 632 ppm.

6.2 Interpretation

The key point is that $Q = 3/2$ is *not an input* in TGP. It is a structural consequence of two features:

1. The exponential dependence of A_{tail} on the eigenvalue index n, which enforces an approximate geometric progression in the mass spectrum.

2. The next-to-leading WKB correction, which tunes the departure from exact geometric progression to the specific value that makes $Q \approx 3/2$.

Neither feature involves any free parameter once g_0^e is fixed. The Koide relation is thus demystified: it is a consequence of WKB quantization in a single-well potential with a particular barrier shape determined by the substrate topology.

This result addresses a long-standing puzzle in particle physics [3, 10]. Previous attempts to derive the Koide relation have invoked discrete symmetries or non-commutative geometry; TGP provides instead a dynamical mechanism rooted in ordinary differential equations.

7 Falsifiable predictions

A scientific theory must make predictions that can be tested and potentially refuted. TGP makes several sharp predictions in the lepton sector.

7.1 Fourth-generation charged lepton

The WKB spectrum (32) does not terminate at $n = 2$. The next soliton excitation, $n = 3$, corresponds to a fourth-generation charged lepton with mass

$$m_4 = m_e \left(\frac{A_{\rm tail}^{(3)}}{A_{\rm tail}^{(0)}} \right)^4 = m_e \, e^{-12\sigma} \, (1 + \delta_3) \, . \tag{39}$$

Extrapolating the corrected WKB series to $n = 3$, the numerical integration yields

$$m_4 \approx 2.4 \text{ GeV} \, . \tag{40}$$

This value is noteworthy for two reasons:

1. It lies below the LEP Z-width exclusion threshold $M_Z/2 \approx 45.6$ GeV, so the Z boson cannot decay into a pair of such leptons. The LEP constraint on the number of light neutrino species ($N_\nu = 2.984 \pm 0.008$) does not apply if $m_4 > M_Z/2$, but since $m_4 = 2.4$ GeV $\ll M_Z/2$, the fourth-generation charged lepton would need to be accompanied by a neutrino heavier than $M_Z/2$ (or have non-standard Z couplings) to evade the LEP N_ν bound. In TGP, the neutrino sector has a different topological structure from the charged-lepton sector, and the fourth neutrino mass can indeed be large.

2. Direct searches at LEP [11] exclude sequential heavy charged leptons with $m_{L^-} < 100.8$ GeV, assuming standard electroweak couplings and a stable or long-lived heavy lepton. However, a 2.4 GeV charged lepton with non-sequential couplings (as predicted by TGP, where the coupling to W/Z is suppressed by the soliton overlap integral) would have evaded these searches if its production cross section is sufficiently small.

The prediction $m_4 \approx 2.4$ GeV is therefore not immediately excluded, but it is testable. Future e^+e^- colliders operating at $\sqrt{s} \gtrsim 5$ GeV with high luminosity could produce such a lepton in pairs and detect it through its decay products.

7.1.1 Kill-shot scenario

If improved direct searches or precision electroweak data conclusively exclude a charged lepton in the range $2 < m_4 < 3$ GeV with any coupling strength, TGP in its current form would be falsified. This is a concrete, near-term test of the theory.

7.2 Cross-sector Koide relations

The same g_0 mechanism predicts that Koide-like relations hold independently in the down-type quark sector (d, s, b) and the up-type quark sector (u, c, t):

$$Q^{(d)} = \frac{(\sqrt{m_d} + \sqrt{m_s} + \sqrt{m_b})^2}{m_d + m_s + m_b}, \qquad Q^{(u)} = \frac{(\sqrt{m_u} + \sqrt{m_c} + \sqrt{m_t})^2}{m_u + m_c + m_t}. \tag{41}$$

These should both lie close to $3/2$, with deviations of order 10^{-3} that are calculable from the respective g_0 values. Current quark-mass uncertainties are too large to test this at the required precision, but lattice QCD determinations are approaching the necessary accuracy.

7.3 Coupling constant correlations

The relation $\alpha_s(M_Z) = N_c^3 g_0^e/(8N_f^2) \times C$ ties the strong coupling to the lepton mass spectrum. Any improvement in the precision of $\alpha_s(M_Z)$ therefore simultaneously tests the lepton mass predictions of TGP. A determination of α_s at the 10^{-4} level would either confirm or refute the TGP relation.

8 Conclusion

We have shown that the Topology-Generated Potential framework reduces the charged-lepton mass hierarchy to a single substrate parameter, $g_0^e = 0.90548$. This parameter, fixed once from the muon-to-electron mass ratio, yields the tau-to-electron mass ratio to 0.016%, the Koide parameter to 0.02%, and the strong coupling constant $\alpha_s(M_Z) = 0.1174$ to within 0.6σ of the PDG world average—all with zero additional free parameters.

The physical mechanism is the exponential sensitivity of soliton tail amplitudes to the substrate boundary condition, combined with the fourth-power mass law $M \propto A_{\text{tail}}^4$ arising from the kinetic coupling $K(\phi) \propto \phi^4$. The Koide relation $Q = 3/2$ emerges naturally from the WKB quantization structure and is not imposed as a constraint.

Limitations and honest framing. Three points warrant explicit statement. First, the third generation is fixed by the Koide closure $Q = 3/2$ at $N = 3$—specifically the WKB relation $y = x^2$ of Eq. (34)—and *not* by a naive golden-ratio ladder: extending the second-generation relation as $g_0^{(\tau)} = \varphi^2 g_0^{(e)}$ (with $\varphi = (1 + \sqrt{5})/2$) overshoots the measured tau by $\sim 13.7\%$ and is therefore not used here. Second, the Brannen rotation parameter $B = \sqrt{2}$ underlying the $Q = 3/2$ value is confirmed *numerically* from the soliton ODE ($|B_{\text{num}} - \sqrt{2}| < 10^{-6}$) but remains *analytically underived* (open problem T-OP3); we report it as a numerically established structural input, not a proven theorem. Third, the *spectral stability* of the second- and third-generation solitons is an open problem (T-OP4): a numerical diagonalization of the fluctuation operator around the soliton profiles shows that, within the *unconstrained* static energy functional whose Euler–Lagrange equation is the soliton ODE, the μ and τ profiles are saddle points (2 and 3 localized negative modes, respectively), while the electron profile carries none. This does not affect the mass ratios reported here (they are properties of the profiles themselves), but full stability presumably requires either the ghost-wall reflection dynamics (currently a non-variational regularization) or a constrained formulation in which the budget of created space is held fixed (Q-ball–type stability); this analysis is deferred to a dedicated study. A first such study (linear constrained analysis, pre-registered criteria) has since been carried out: *linear* budget-type constraints (four families and their pairs) do *not* restore stability—the minimum attainable saddle index is 1 for both μ and τ, and the surviving mode is not the profile-family direction (overlap < 0.01)—so the constrained route remains open only at the nonlinear (charge-type) level. The same study finds the mass-ratio mechanism sensitive to the wall model: within a smooth regularization family f_ε the τ profile ceases to exist (collapse) for every ε tested, and the μ-based r_{21} drifts by $+1.9\%$ to $+23\%$ against the hard-wall baseline; the hard-wall reflection therefore remains a structural input of the present mechanism rather than a removable regularization.

The theory makes a sharp falsifiable prediction: a fourth-generation charged lepton at $m_4 \approx 2.4$ GeV. Detection or exclusion of this state at future collider experiments will provide a decisive test of TGP.

The results presented here represent one sector of a broader programme. The full TGP manuscript [6] derives the complete fermion mass matrix, the gauge coupling unification conditions, and the cosmological constant from the same substrate topology. The N-body soliton dynamics, including confinement and hadron spectroscopy, are treated in a companion paper [12].

Acknowledgments

The numerical calculations were performed using custom Python code available in the TGP repository [7].

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